Collaborative Intrusion Detection System for SDVN: A Fairness Federated Deep Learning Approach

Jie Cui, *Senior Member, IEEE*, Hu Sun, Hong Zhong, Jing Zhang, Lu Wei, Irina Bolodurina, and Debiao He

***Abstract*—With the continuous innovations and development in communication technology and intelligent transportation systems, a new generation of vehicular ad hoc networks (VANETs) has become increasingly popular, making VANET communication security increasingly important. An intrusion detection system (IDS) is an important tool for detecting network attacks and is an effective means of improving network security. However, existing IDSs encounter several problems involving inaccurate detections, low detection efficiencies, and incomplete detections owing to extensive changes in vehicle locations in VANETs. This study explores federated learning in software-defined VANETs and designs an efficient and accurate collaborative intrusion detection system (CIDS) model. The model utilizes the collaboration among local software-defined networks (SDNs) to jointly train the CIDS model without directly exchanging local network data flows to improve the expansibility and globality of IDSs. To reduce the model difference between different SDN clients and improve the detection accuracy, this study regards the prediction loss for each SDN client as an objective from the perspective of constrained multi-objective optimization. By optimizing a surrogate maximum function containing all the objectives, the method adopts two-stage gradient optimization to achieve Pareto optimality for SDN clients with the worst fairness constraint maximization performance. In addition, this study evaluates the training model using two open-source datasets and compares it with the latest methods. Experimental results reveal that the proposed model ensures local data privacy and demonstrates high accuracy and efficiency in detecting attacks and is thus superior to the current schemes.**

***Index Terms*—Federated deep learning, collaborative intrusion detection system, intelligent transportation system, convolutional neural network, gradient optimization.**

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I. Introduction

In recent years, the mobile communication technology has continuously developed and upgraded. As new types of mobile communication networks, 5G networks have the characteristics of higher speeds and lower delays compared to previous generation networks; thus, 5G networks are capable of satisfying the requirements of Internet of Things (IoT) applications and have emerged as the key infrastructure supporting economic and social digitization, networking, and intelligent transformation [1]. Currently, the new generation of intelligent transportation systems (ITSs) is in the final stage of large-scale deployment worldwide. Relying on the advantages of the new generation of mobile communication networks, new intelligent vehicles play an increasingly important role in ITSs through vehicular ad hoc networks (VANETs), which are dynamically composed of vehicles, roadside units (RSUs), base stations, cloud servers, and trust institutions. VANETs not only improve road congestion and relieve traffic pressure but also ensure traffic safety. Onboard communication technologies can be divided into two categories: dedicated short-range communication and long-term evolution (LTE). According to the characteristics of the new generation 5G network communication, vehicle-to-everything communication based on LTE can support various vehicle communication services for safe and non-safe applications by using high capacity, large and small area coverage, and widely deployed infrastructures. However, owing to the low delay and other characteristics of current VANET communication, several challenges such as security, privacy, and quality of service supply still need to be addressed in the network [2].

Without an appropriate authentication protocol in a VANET, an attacker may inject unauthenticated messages to gain personal benefits [3]. For example, an attacker can reduce the traffic volume on a target road by misleading other vehicles in the agent area, thereby broadcasting false traffic congestion messages to reduce road congestion. Thus, communication security is a major challenge for new generation vehicle networks. If the security of a communication network is not strengthened, it may lead to the collapse of the entire communication network and may even affect the entire ITS. VANETs are vulnerable to several

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attacks including spoofing, denial of service, and message delay attacks. In that regard, an intrusion detection system (IDS) is an effective method for actively responding to such network attack threats and has been widely used in network anomaly detection [4]; it is capable of detecting malicious behavior by monitoring the host or network to provide network security. According to the classification of architectures, existing IDSs can be categorized into three types: centralized, hierarchical, and collaborative IDSs.

Currently, most IDSs are deployed in different locations, like on the vehicle, cluster head or RSU, which can affect the performance of the IDSs [5]. However, majority of these solutions are partly based on the whole VANET, are trained by artificial intelligence methods, and are limited to all abnormal network behaviors detected under their local subnetwork rather than the entire vehicle network. A collaborative intrusion detection system (CIDS) can address this shortcoming of local IDSs and is defined as a collection of host-based or network-based intrusion detection systems. A CIDS allows the sharing of information and detection of network attacks in a vehicle network. A classic CIDS unit is composed of local monitoring, global monitoring, association and aggregation, and data-publishing components [6]. CIDSs are typically developed on the basis of distributed machine learning (DML) for detecting known and unknown attacks with some generalization capability [7]. The DML algorithm is a multi-node training-based machine learning algorithm that aims to enhance the efficiency, accuracy and scalability of computation on large-scale data sets and address the limited computational power of a single entity.

A key security requirement of a CIDS based on DML is to promote sharing among machine learning models without compromising data privacy. In particular, DML is susceptible to the attacks of model inversion, wherein adversaries can deduce the sensitive training data from the learning results [8]. The federated learning (FL) model guarantees the privacy of ordinary DML algorithms. FL [9] is a distributed algorithm based on a central server, which is different from the ordinary DML algorithm. In terms of its topology, FL typically adopts a star topology, whereas DML typically adopts a point-to-point topology; it is capable of learning from distributed data sources without having to access original data samples across different data sources and can learn models from scattered data without sacrificing data privacy. Given that the data from different client data sources are inconsistent, FL approaches primarily focus on improving the algorithm performance and reducing algorithm differences. For FL, algorithmic differences are reflected in two aspects, the first one is brought by the difference of dataset distribution (mainly the number of datasets, data features and label inconsistency), which leads to the differences of FL clients' training models; the second one is the differences of clients' training models, which leads to the differences of FL server aggregation algorithms. Due to the privacy of FL client's private dataset, the currently FL mainly focuses on reducing algorithmic differences from the latter aspect.

Notably, most FL models are based on the two-stage method of vehicles and RSUs, vehicle participation in training, and the RSU aggregation model. However, this leads to a huge computational burden on RSUs, and malicious vehicles may participate in training. Software-defined networking (SDN) is a carrier that improves the communication efficiency and traffic forwarding of VANETs, which can prevent vehicles from directly participating in the training. SDN is an effective approach for providing security-control flexibility and programmability for networks. With a logic centralized control plane and programmability, an SDN controller can provide a good means of improving network security [10]. An SDN is composed of a control plane, a data plane, and an application plane. The control and data planes are decoupled from each other to logically centralize the network state and intelligence [11]; however, the mode of the central controller may become a bottleneck in the system and is prone to single-point failure [12], which poses a significant security threat to ITSs. To alleviate scalability and reliability problems, distributed SDNs have been developed by distributing and sharing workloads among multiple SDN controllers [13]. The entire network is composed of multiple subnetworks, each of which is managed by an independent SDN controller.

With the increasing number of vehicles in VANETs, network communication and data exchange are becoming more and more frequent. SDN can provide security control to the network and relieve the pressure of communication management in VANETs. Meanwhile, SDN controller can train intrusion detection models and SDN application plane can deploy intrusion detection systems that can detect network anomalies while managing network flows. Most existing schemes use vehicles as training entities in VANETs, but vehicles have high mobility and limited computational resources, and these schemes may be a little inappropriate. The introduction of SDN in VANETs is beneficial to avoid threat vehicles joining the training directly when training the model and generating insecure detection models; it is beneficial to the management of vehicle edge network and reduces the pressure on vehicle edge computing resources in VANETs; it is beneficial to adjust the deployment of VANETs scenarios and improve the scalability. If only a single SDN is deployed in VANET, all network traffic is sent to the cloud server and SDN for training and detection, which brings huge computation and communication costs to the system. However, by training and deploying IDS on multiple distributed SDNs, it can detect abnormal behavior of VANET network traffic within the management control of SDNs without incurring huge overhead. Therefore, it is important to design an efficient intelligent intrusion detection method based on the entire vehicular network.

This paper proposes a new software-defined VANET (SDVN) intrusion detection framework that combines FL and SDNs training detection model with the security model of SDVN to construct an efficient and accurate collaborative intrusion detection model ensuring the security of ITSs. In the proposed CIDS, multiple SDN clients train models under their management subnetworks and upload them to the centralized cloud server to complete the model aggregation task. The cloud server sends the global aggregation model (including the weights and gradients) to each SDN client. The intrusion detection model deployed by the SDN application plane can detect abnormal VANET

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behavior under its management subnet. The main contributions of this study are as follows:

- This study utilizes FL and a distributed SDN controller to develop a new collaborative intrusion detection framework in the SDVN and construct a training network model using a deep learning convolutional neural network, which improves detection performance, ensures data privacy, and avoids high system overheads.
- A two-stage gradient optimization is utilized from the perspective of multi-objective optimization. Considering the prediction loss of the client as the target, the surrogate maximum function including all the targets is used to optimize the target based on the gradient in the client with the worst fairness constraint maximization performance to achieve Pareto optimization; this reduces the differences in the performance of different client detection models.
- Experiments are conducted on independent and identical distribution (IID) and non-IID datasets, and the proposed method is compared with some existing schemes, both of which have better performance.

The remainder of this paper is structured as follows. Section II introduces related research. Section III presents the system model and its composition. Section IV introduces the proposed collaborative intrusion detection architecture and the functions of each module. Section V introduces comparative and validation experiments. Finally, Section VI summarizes this study.

II. Related Work

In this section, we briefly summarize the existing intrusion detection models proposed by researchers on deep learning (DL), federated learning, and machine learning (ML).

Abou El Houda et al. [14] proposed a novel centralized framework based on Explainable Artificial Intelligence, which relies on novel IDS for IoT networks, and develops through deep neural networks for detecting IoT-related intrusions. Moreover, providing cybersecurity experts with an explanation of any critical decisions that improves the credibility and security of the model. Otoum et al. [15] proposed a constrained Boltzmann machine cluster intrusion detection system to detect known and unknown attacks based on a wireless sensor network (WSN), which is a potential DL-based IDS methodology. Javaid et al. [16] proposed a network anomaly intrusion detection system that is based on a deep learning approach, which focuses on the self-learning and Soft-max regression algorithms of sparse autoencoders. However, they take into account only the intrusion detection algorithms in a singular system. That means, all the intrusion detection data are centralized on a singular device or system. Li et al. [17] proposed a collaborative intrusion detection system based on semi-supervised learning which solves the problem of very limited data labels and also improves the performance of a single detector in the Internet of Things. Wahab et al. [18] proposed an intelligent detection model based on multi-decision support vector machine (SVM) in VANET, that reduces the communication overhead and improves the detection rate.

A VANET is a dynamic network with large coverage. Training the detector only for the network flow collected in a small area will lead to biased flow and generalization problems. Alcoy et al. [19] proposed that a DoS attack is a major challenge to network security, especially in a VANET. Traditional independent IDS is difficult to effectively prevent such attacks, but collaborative intrusion detection systems can effectively prevent or detect DoS attacks.

Therefore, several devices that have intrusion detection data can work collaboratively to accomplish the construction of an intrusion detection model rather than having to build training data on a singular device. Nguyen et al. [20] proposed a decentralized anomaly detection system that is based on IoT devices. By building network communications between devices, intrusion detection can effectively detect IoT devices that are subject to security threats. Due to its centralized aggregation, the model of the central server is potentially manipulated by the adversaries. Tian et al. [21] proposed an RSU-based split-level intrusion detection system, which transmits raw network traffic to RSUs for training. A trained model can monitor the whole vehicle network by detecting the network of control information with packets in real time. Sedjelmaci and Senouci [22] proposed an accurate and light intensive intrusion detection architecture for VANETs based on clustering algorithms that cope with high mobility of nodes and rapid changes in topology. Aloqaily et al. [23] conceived an intrusion detection scheme to combat the security attacks in VANET by utilizing deep belief with decision trees. We understand that when these approaches are deployed on RSUs, they are prone to a single point of failure as well as being poorly scalable, requiring huge storage and computational capacity in the RSU.

Shu et al. [24] discussed distributed SDN controllers and used generative adversarial networks (GAN) of deep learning to devise a collaborative intrusion detection system in VANET, which solves the biased flow problem of the single detector and reduces the high cost of centralized detectors. However, during the training process, training parameters are directly exchanged between SDNs, which lacks security. In terms of security, the IDS based on federated learning can guarantee the privacy of training. Preuveneers et al. [25] built a blockchain-based anomaly detection model where multiple intrusion detection models establish a distributed ledger through federated learning to achieve each intrusion detection model's traceability, thereby preventing malicious manipulation and poisoning attacks. Based on trusted federated learning and blockchain, Lu et al. [26] designed a collaborative data share architecture. Through building a decentralized collaborative machine learning system among several devices, it ensures the accuracy and security of the aggregate machine models. The privacy protection is enhanced by integrating differential privacy into federated learning in order to address the speculative attacks and privacy leaks associated with data sharing. Liu et al. [27] proposed a cooperative network intrusion detection model based on vehicles and RSUs, and saved the distributed training model to the blockchain to realize sharing.

Most of the current study only considers the use of the algorithm itself and rarely considers the optimization of the algorithm. Carvalho et al. [28] utilized an ant colony optimization algorithm to improve the accuracy of the machine learning intrusion detection model. Ghada Abdelmoumin et al. [29]

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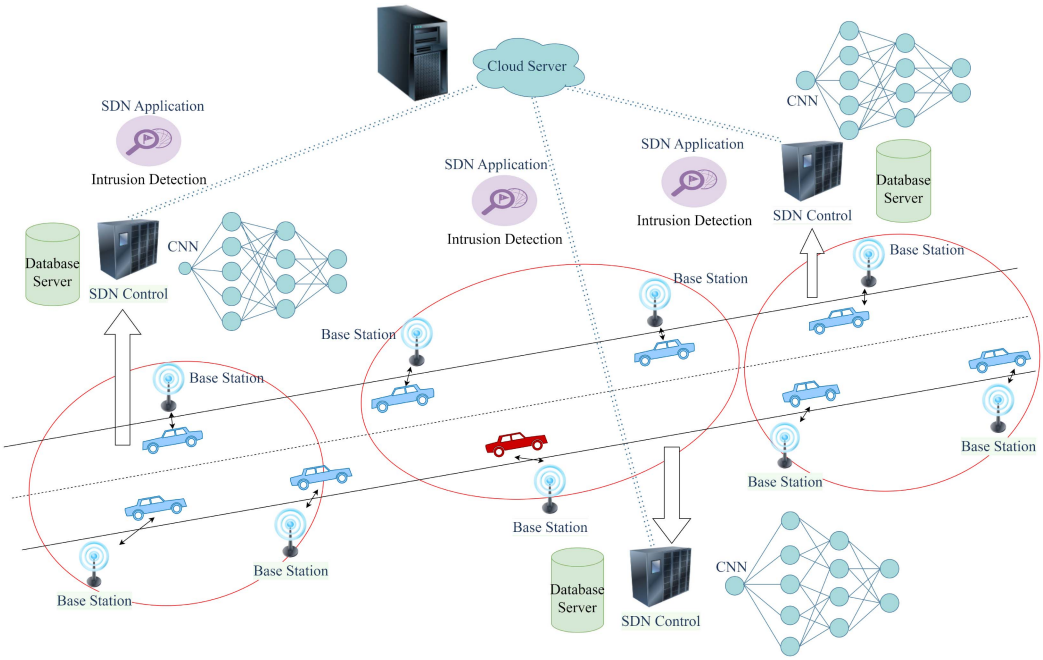

Fig. 1. New deep learning collaborative intrusion detection model for SDVN.

researched anomaly-based machine learning-enabled intrusion detection systems (AML-IDS) and proposed two optimization methods (hyperparameter tuning and ensemble learning) to optimize and evaluate AML-IDS in IoT environment dataset in response to the problem that AML-IDS performance and prediction accuracy are lower than DL-IDS. Raja et al. [30] proposed a cooperative intrusion detection system based on the distributed machine learning algorithm of the alternating direction multiplier method (ADMM), in which all vehicles cooperate to create a global classifier and add a differential privacy paradigm, which ensures the security of training process parameters.

III. System Model

In this section, we will introduce the collaborative intrusion detection system model proposed in this study, which is shown in Fig. 1.

A. The Proposed Intrusion Detection System Model

The collaborative intrusion detection system model proposed by us is composed of cloud servers, multiple SDN controllers, base stations, and vehicles. In this paper, it is assumed that there is one cloud server and three SDN controllers to build a distributed federated deep learning model. The main symbols in this study are as shown in Table I.

Cloud Server: The cloud server obtains the real time system status from multiple distributed SDN controllers, provides services for the whole vehicle network, and improves the data analysis and calculation power of intelligent transportation. As the central server for aggregating multiple client model parameters and gradients in the federated learning algorithm, it mainly distributes the global model parameters, gradients, and weights to the SDN controller to protect the data privacy of the SDN client.

TABLE I
Symbols and Definition in the Paper

Symbol	Definition
N	number of SDN client
S_k	k-th SDN client
S	SDN client sets
ℓ_k	model utility loss of the k-th SDN client
L	training loss of deep learning model
D_k	dataset resources on the k-th SDN client
$H(\theta)$	global model with parameter θ
R	communication rounds
E	training epoch
B	training batch size
T	number of training batches
ε	fairness budget
W	weight parameter
m, n	first momentum, second momentum
η	learning rate
u, v, σ	deep learning optimizer training parameter constant
x, y	sample features, sample label
$\hat{y}$	predicted label

SDN: The SDN data plane is composed of base stations within the management subnet of SDN and the vehicles in the area to realize the network communication transmission and data exchange in VANET; The SDN control plane is responsible for the control of base stations data plane within the subnet and the real-time detection of base stations network data within its management area. In the training process, the SDN controller

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serves as the client of the federated learning algorithm training. The federated learning algorithm extracts network traffic information dataset after normalization detection from the local database server for training, and the dataset would not be shared; The SDN application plane deploys the well-trained intrusion detection model, which facilitates real-time training and improves the efficiency of detection. During detection, the network traffic information of local vehicle communication is extracted from the base stations, and the deployed detection model is employed to detect network attacks. In addition, to facilitate periodic updates of the intrusion detection model in VANETs, each SDN is configured with a database server to store the dataset of detected network traffic, which provides the basis for the next training model.

Base Station: As an RSU of a VANET, which is used to realize wireless communication in vehicle-to-vehicle (V2V) and vehicle-to-network (V2N). Several base stations can form a local subnet managed by SDN controller, and the number of base stations can be increased according to the needs of the local subnet, which has scalability.

Vehicle: There are onboard units and application units in the vehicle, which use wireless cellular network communication with external communication, and base stations as the carrier of communication to spread significant information, including speed, location, and road conditions.

The proposed collaborative system model is deployed in the SDN application plane, which can improve detection efficiency and increase scalability. In the detection process, first multiple SDN application planes deploy training models, then multiple SDN controllers extract base station and vehicle communication network data, and parse, truncate/fill and normalize the original packets, then detect this network data by the intrusion detection model deployed in the SDN application plane, and after detection, label this normalized data and store it in the corresponding database server. During the training process, multiple SDN controllers establish a secure and reliable data share channel with local database servers to ensure the privacy of the transmitted training data, and SDN controllers directly read private data sets from the corresponding database servers to train local models; second, multiple SDN controllers train local detection models through deep learning convolutional neural networks, and upload information such as model parameters and gradients; Finally, the aggregated and optimized models are sent to multiple SDN controllers, and the models are deployed to the SDN application plane to detect the abnormal communication networks behaviors in VANETs.

In our proposed collaborative intrusion detection model, an efficient and accurate detection model is trained using a federated learning algorithm that can be independent of the distribution of datasets, i.e., it can be IID or non-IID. The real-world vehicular network environment is complex and the distribution of datasets in the subnets managed by the SDN controller is also non-IID and homogeneous, and most current federated learning algorithms [24], [25], [26] adopt a federated average algorithm for cloud server integration, so in order to reduce the impact of non-IID datasets on the performance of the training model, we utilize a gradient-based two-stage optimization algorithm from a multi-objective optimization perspective for cloud server integration optimization. In the first stage, we first implement a fair min-max model by using multi-client fairness; in the second stage, we constrain Pareto optimization to achieve a Pareto optimal model. The objective of the gradient-based two-stage optimization can be defined as:

$$\min[\ell_1(H), \ell_2(H), \ldots, \ell_N(H)], \tag{1}$$

$$\text{s.t. } g_k(H) - \varepsilon_k \leq 0, \forall k \epsilon \{1, 2, 3, \ldots, N\}. \tag{2}$$

where (1) is the objective function, (2) is the constraint, k represents the k-th client, ℓ represents the model utility loss, H represents the global model, g_k represents the performance difference between the current model of the k-th client and the global model, and ε_k represents the difference between the minimum-maximum model performance, and only if the former is less than or equal to the latter, the performance of each client is guaranteed to be increasingly better.

Concerning data privacy, federated learning is privacy assured and can protect the security and privacy of local client training data. In terms of SDN controller training dataset collection, our proposed framework is also able to ensure privacy security. First, the base station is secure and trusted, so the data source is secure; second, the SDN controller can control the network transmission within the data plane composed of the base station and its internal vehicles, and the network transmission is protected by network security protocols; third, if a threatening behavior is generated by an intruder to the process of collecting data, the detection model deployed in the SDN application plane can detect the anomalous behavior, which also ensures privacy security to a certain extent; fourth, the SDN controller is capable of detecting the anomalous behavior in the SDN application plane. Fourth, the data trained by the SDN controller will first be normalized and labeled, and the pre-deployed detection model can identify threatening data.

The trust mechanism management in SDVN environment can improve the detection rate of attacks to some extent and also reduce the risk of adversary attacks. Abdel-Basset et al. [31] improve the detection of malicious vehicles by learning the spatial-temporal representations of vehicles through Transformer networks, specifically location and time features. In the proposed system model, the distributed SDNs manage the vehicles within its subnetwork and the vehicles send packets with spatial-temporal representations, i.e., characteristics such as vehicle location, time, passed base stations and its information. Gao et al. [32] proposed a direct trust management approach based on historical interaction and a recommended trust approach with Bayesian inference. In our model, the presence of an intrusion detection model and database server deployed in the SDN application plane can record vehicle historical interaction information, and once anomalies are detected, they are reported to the trust authority in VANETs that holds the vehicle accountable and adjusts that trust value.

B. Threat Model

In the traditional federated learning framework, the central server and client training show a high degree of cooperation. The vehicles (as the client) and RSU (as the central server) participate

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in the training, which can train an efficient intrusion detection model. However, the traditional federated learning centralized intrusion detection model is vulnerable to the following security threats:

Inference Attack: A central server takes charge of aggregating all clients' information on local models, attacker launches an attack on the central server to infer local client training data, violates clients' data privacy, and thus attacks VANET.

Single-Point-of-Failure (SPoF) Attack: When a single point of failure occurs on central server, the global model can't be aggregated. An attacker, through an edge client, participates in training to launch a poisoning attack and destroy the global model.

Byzantine Attack: The attacker client can randomly pass misleading or low-quality local parameters, thus degrading the performance of the aggregation model of the central server, and eventually disrupting the availability of the global model.

Sybil Attack: The attacker client spoofs the central server, resulting in numerous fake client identities sending false messages to the central server.

Label Flipping Attack: While training an intrusion detection model, a malicious client may vary the local training labels to provide poor quality and inefficient model updates, thus affecting the overall learning and training. For example, a vehicle may send malicious labels at a certain time interval, and this will have an impact on the detection model's effectiveness.

Data Poisoning Attack: During training, the attacker client injects strong noise to corrupt the quality of the local training data and provide incorrect local gradient information.

Given the communication security of VANETs, the enemy may also trigger different attacks, leading to the collapse in VANETs:

Denial of Service (DoS) Attack: In VANETs, network resources are usually supposed to be limited and when other vehicle nodes send vital data to other vehicle nodes, DoS attacks interfere with the work in VANETs by jamming the communication channel to deny legitimate vehicles access to the network and its resources.

Distributed Denial of Service (DDoS) Attack: Distributed denial-of-service attacks are more severe and more powerful than traditional DoS attacks. In this variant of DoS attack, instead of a single attacker, multiple nodes act as attackers to overwhelm the nodes.

Wormhole Attack: The attack normally associates with two or more malicious nodes and sends packets from one side of the malicious node to the other node through a tunnel.

Spoofing Attack: The attacker pretends to be a legitimate node to steal sensitive and confidential information and gain super privileges. One of the attacks is an in-between attack, located between two communicating nodes.

In response to the above attacks, to solve the threat of traditional federated learning intrusion detection model and a VANET security, we propose a federated learning intrusion detection model based on SDN and Cloud Server, which avoids the possible threat of vehicles directly participating in training as clients, thus improving the security of the training intrusion detection model. At the same time, both SDN and Cloud Server deploy intrusion detection models to ensure the security in VANETs communication and avoid third-party attacks on training model entities in VANETs. Specifically, the CIDS of CNN-based FL is a model trained from a public network security dataset, so the objects that the training model can detect are also specific network datasets, and what the model identifies is data. When the attacker launches an attack, the carriers are all network communication messages, and the detection model can detect the network attack. For the CIDS model, the attacker initiates the attack through network communication, generating network communication packets, and then the packets are parsed and normalized into data that can be recognized by the model. The detection model extracts data features based on parameters such as the trained convolution kernel, and finally by determining whether it is an attack or not. If it is recognized as an attack, it immediately reports to the trust institution of VANET to hold the attacker accountable for the attack and stop it from further threat to VANET security.

IV. Collaborative Intrusion Detection Model

In this section, we first outline the workflow of the federated learning scheme, then introduce the designed deep learning intrusion detection model based on convolutional neural network (CNN), and finally the proposed federated learning scheme in detail and introduce the two-stage gradient optimization scheme based on federated learning.

A. Federated Deep Learning Intrusion Detection System Process

The basic idea of the scheme is to build an SDVN collaborative intrusion detection architecture, which means that multiple SDNs are integrated into the VANET to manage the roadside base stations under their sub-networks and build a federated deep learning intrusion detection model together with the cloud server. That model is based on a federated learning framework and a two-stage gradient optimization algorithm based on multi-objective optimization, and the process is as follows (as shown in Algorithm 1).

1) System initialization: In the initialization phase, the cloud server pre-trains the model for the intrusion detection model based on deep learning, selects a set of initialization parameters W^0 and some other parameters related to the model training, namely the learning rate η, Loss function L, local epoch E, batch size B and communication rounds R.

2) The SDN client trains the local model: After receiving initial model parameters, η, L, B, E and R from the cloud server. each SDN client uses the data resources D_k in the local base station to locally train an intrusion detection model based on deep learning CNN. Algorithm 2 summarizes the detailed training process. After the SDN client, the deep learning model is trained, and the parameters are uploaded to the cloud server.

3) Cloud server parameter aggregation and optimization: After receiving each SDN client deep learning model parameter, the cloud server aggregates the model parameters uploaded by all SDN controller through the aggregation function $Aggregate()$.

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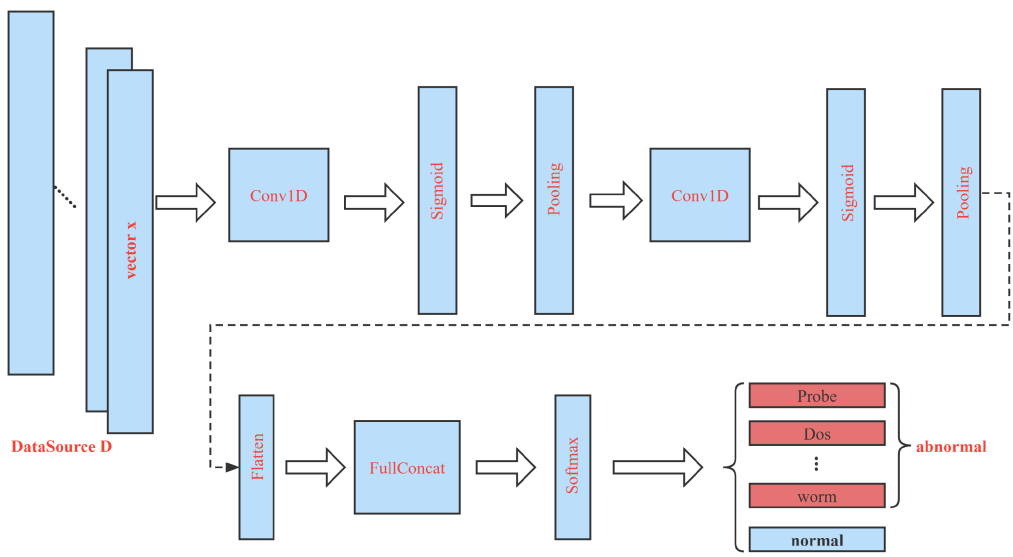

Fig. 2. Convolutional neural network module for local SDN client.

First, fairness is regarded as a constraint, and a multi-objective framework optimization with multiple fairness constraints is optimized for the model aggregation parameters while maintaining the utility of the model to the maximum. Second, the optimized parameters $W_k^{r''}$ are sent back to the SDN client (as shown in Algorithm 3).

4) SDN client local model update: The model aggregation optimization parameters from the cloud server are received, and then the deep learning model of the local SDN client is updated. After multiple rounds of training interaction iterations between the cloud server and all SDN clients, all SDN clients can finally obtain a comprehensive and efficient intrusion detection model which is based on deep learning.

B. A CNN-Based Local Intrusion Detection

In this part, we describe in detail the convolutional neural network algorithm model components and the training procedures. For the traditional classifier model, which is simple and suitable for small datasets and achieves good classification performance, but the limitation is that cannot handle large datasets, we use the datasets, such as public network security KDD, five million network flow data, not only a large amount of data, and more features, so not the traditional classifier model. For example, SVM algorithm, the model is simple, but when the data amount is large, using the kernel function calculation amount is huge. For Deep Neural Networks (DNN) model, it is a multi-layer perceptron model. Since DNN adopts the form of full connection, the more features in the dataset, the more weight parameters, which is not only not easy to fit, but also easy to fall into the local optimum and deviate from the true global optimum. Illy et al. [33] pointed out the shortage of the single DNN deep learning model, which is difficult to handle the complex unbalanced data distribution. Therefore, a hybrid multi-stage DNN intrusion detection framework is proposed, where multi-stage DNN models collaboratively train models. The classification model is trained using DNN models in multiple stages, and the training set used in each stage is derived from the data misclassified in the previous stage until it converges and is less than the prediction threshold. If a hybrid multi-stage deep learning model is used in our scheme, considering the application scenario of VANETs with a large number of vehicles with high mobility, extremely unbalanced data distribution as well as extraordinary differences, it is difficult for the model to converge and meet the demand to some extent. CNN model can solve the problem of DNN quantity expansion of parameters and extract features by sharing convolution kernel. In addition, the CNN model has low complexity compared to other deep learning algorithms and has great advantages in processing large datasets, high prediction accuracy, and reliable results.

1) Model Components: The proposed model primarily consists of a convolutional neural network module, a fully connected layer and a softmax layer (as shown in Fig. 2 *CNN module*). They are described as follows:

CNN Module: There are two convolutional blocks, each with one convolutional layer, sigmoid layer, and max pooling layer. The one-dimensional vector x in the data resources is converted into a column vector, and the data features are extracted by 1D convolution ((3) and (6)) through the double CNN module and normalized by sigmoid (4). The purpose is to ensure that our feature values are within a small range. Then, the maximum value of the features is extracted by the maximum pooling layer (5) as the local feature. The purpose of using the double CNN module is that the network data flow features are many and complex, and the double CNN module can maximize the extraction of features.

$$h_1 = Conv1D_1(x), \tag{3}$$

$$g_1 = Sigmoid_1(h_1), \tag{4}$$

$$x_1 = Maxpooling_1(g_1), \tag{5}$$

$$h_2 = Conv1D_2(x_1), \tag{6}$$

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Algorithm 1: Fairness Federated Deep Learning Algorithm.

Input: SDN client set S, Data set of all SDN clients $\{D_k | k \in \mathbf{N}\}$, number of communication rounds R, fairness budget ε.
Output: Federated deep learning intrusion detection model
Initialiation:
(a) The cloud server initializes the model parameters W^0, η, L, B, R and E;
(b) Each SDN client S_k initializes a training set D_k and initializes fair budget parameters ε;
(c) The SDN client S_k initializes training epochs E and communication rounds index by $r = 1$;
Procedure:
for $r \leq R$ **do**
(1). **For SDN clients** S **:**
for $\forall k \in N$ **do**
S_k calculate the local model parameter W_k^r of the r-th training using the input parameter W^{r-1}, η, L, B, D_k and E according to Algorithm 2.
for $\forall j \in T$ **do**
S_k calculate the local model parameter $W_{k,j}^r$ of each batchsize B.
end
S_k upload model parameters $\{W_{k,j}^r | j \in T\}$ to the cloud server.
end
(2). **For cloud server :**
for $\forall k \in N$ **do**
$W_k^{r'} =$ **Aggregate**$(W_{k,1}^r, W_{k,2}^r, \ldots, W_{k,j}^r)$
The cloud server optimizes the local model parameter $W_k^{r'}$ to $W_k^{r''}$ of each SDN client according to Algorithm 3.
end
The cloud server sends the optimized parameters $W_k^{r''}$ to all SDN clients S_k.
(3). **For SDN clients** S **:**
for $\forall k \in N$ **do**
The SDN client S_k uses $W_k^{r''}$ to update its local deep learning detection model parameters.
end
$r \longleftarrow r + 1$
end
return A deep learning model with well trained parameters deployed on the SDN client.

$$g_2 = Sigmoid_2(h_2), \tag{7}$$

$$x_2 = Maxpooling_2(g_2). \tag{8}$$

Full Connection Layer: After the network data flow passes through the convolution layer, many data features will be obtained. The full connection layer is used to combine these features to classify (10) and then determine whether it is an abnormal network data flow.

$$x_3 = Flatten(x_2), \tag{9}$$

$$x_4 = FullConcat(x_3). \tag{10}$$

Softmax Layer: A softmax layer is used for mapping the output to the probability distribution of the prediction class (11). Given a network data stream feature vector x (one-dimensional vector) as the input of the model, the CNN module processes it to exact features and finally judges whether the network data is abnormal through the full connection layer and the softmax layer.

$$\hat{y} = Softmax(x_4). \tag{11}$$

Logistic regression loss function: N represents the number of SDN clients, B represents the batch size of training samples, y represents the actual sample label, and $\hat{y}$ represents the predicted sample label.

$$L = -\frac{1}{B} \sum_{j=0}^{B-1} \sum_{i=0}^{N-1} y_{i,j} \ln \hat{y}_{i,j} + (1 - y_{i,j}) \ln (1 - \hat{y}_{i,j}). \tag{12}$$

2) Local deep learning model training

Each SDN client S_k refers to Algorithm 2 and trains a local model according to the proposed CNN-based deep learning model on its dataset resource D_k. Specifically, in the training process of each SDN client, each SDN client S_k first updates the training initialization model parameters according to the given update model parameters W_k^r. Then, the SDN client retrains

Algorithm 2: Local Model Training Based on DL-CNN.

Input: $\eta, \mu, \nu, \sigma, B, L, D_k, W_k^{r-1}$
Output: W_k^r
Initialiation:
(a) Initialize the first and second momentum variables by $m_0 = 0, n_0 = 0$ and $t = 0$;
(b) Set the model parameters by $W_k^r \longleftarrow W_k^{r-1}$;
(c) Split D_k into batches with equal size B;
Procedure:
repeat
for $\forall batch \in D_k$ **do**
for $t \leq T$ **do**
(a) Compute the gradient by $g_t \longleftarrow \bigtriangledown W_k^r L$;
(b) Update the biased first momentum estimate by $m_t = \mu * m_{t-1} + (1 - \mu) * g_t$;
(c) Update the biased second momentum estimate by $n_t = \nu * n_{t-1} + (1 - \nu) * g_t^2$;
(d) Update the biased-corrected first momentum estimate by $\hat{m}_t = \frac{m_t}{1-\mu^t}$;
(e) Update the biased-corrected second momentum estimate by $\hat{n}_t = \frac{n_t}{1-\nu^t}$;
(f) Update the model parameters by $W_k^r = W_k^r - \frac{\hat{m}_t}{\sqrt{\hat{n}}+\sigma} * \eta$;
end
end
until The loss function L minimizes;
return W_k^r

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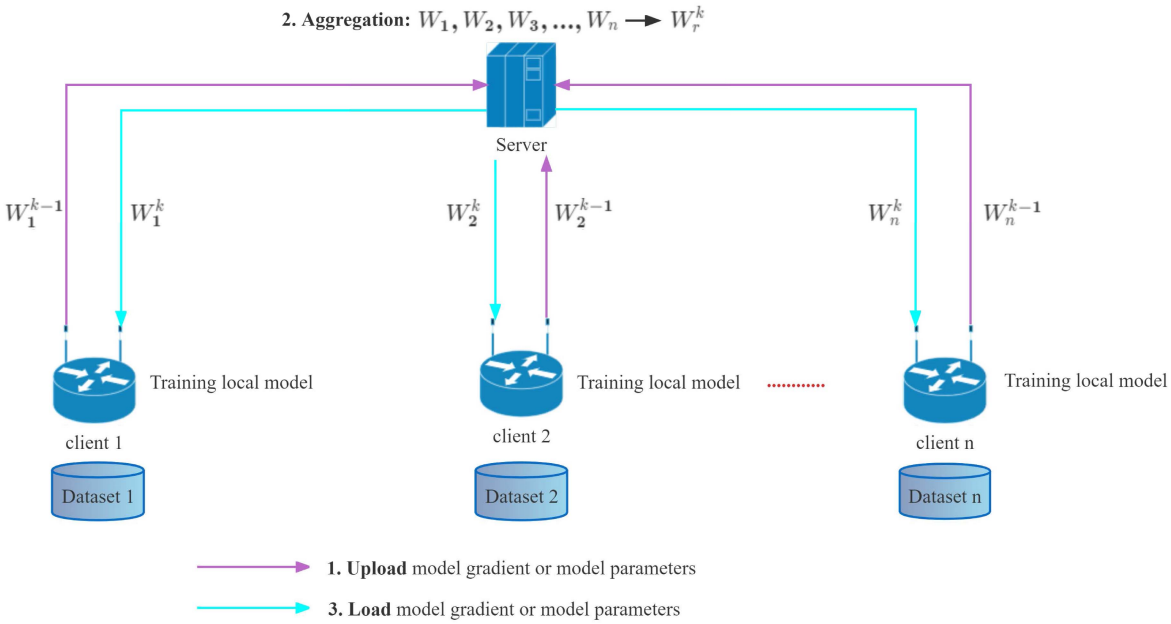

Fig. 3. Server and client aggregation optimization model.

the deep learning model using the same dataset resource D_k, uses the Adam optimizer to promote the rapid convergence of the loss function, and uses the SGD optimizer to slowly find the optimal solution. In Fig. 2, there are two major categories of model categorization output data, normal and abnormal, and there are multiple types of attack data in the abnormal category, mainly four major abnormal categories, such as DoS Attack, Probing Attack, Remote to Local Attack, and User to Root Attack. Within each large attack class, there are multiple labels for smaller attack types. These abnormal type of data are closely related to the threat model. For example, Smurf, as a small label type of attack, sends a large number of IP packets, exhausts bandwidth, disrupts server availability, and generates DoS attack model.

C. Federated Deep Learning

Federated learning is a kind of distributed learning algorithm, which can learn and train from distributed data sources without accessing original data samples across different data sources. Liu et al. [34] proposed a hierarchical federated learning (HFL) model, client-edge-cloud, in which client trains the model, edge server aggregates the local model, and cloud server aggregates the edge model. This design enables to reduce the communication overhead and simplifies the management of vehicular network, but given the intrusion detection model of vehicular network environment, it is unreliable for vehicles to train the local model directly, which is reflected in the possible existence of adversary vehicles in the vehicles to attack the system during the training process. Unlike traditional FL based on cloud, HFL needs to consider not only the non-IID of client datasets but also the non-IID among edge servers. In addition, the aggregation model twice in the cloud and edge generates huge computational overhead. Thus, we choose the traditional Cloud-based FL architecture, i.e., client-cloud. In SDVN, multiple SDN controllers act as clients and a cloud server acts as FL central server. The main description is as follows (as shown in Fig. 3).

Client: N SDN modules form a distributed structure to protect the safety of a VANET under its management subnet. The SDN data plane is the base station, the SDN controller trains the deep learning model, the SDN application plane deploys the intrusion detection model, and the private dataset trained is in the local database of the base station.

Server: The central server of VANETs aggregates and optimizes the local deep learning model parameters of the client.

Suppose that there are N SDN clients $(S_1, S_2, \ldots, S_N,)$, and each SDN client has a specific dataset $D_k = \{X, Y\}$, where X represents the input characteristics and Y represents the output labels. The k-th SDN client s has n^k samples, and each sample is represented as $\{x_i^k, y_i^k\}$. The goal of federated learning is to build a global model H by collaborative learning global parameters and predict the label $\hat{y}$ on each SDN client. The classical federated learning is intended to minimize the experiential risk for all SDN client sample data, namely:

$$\min_{\theta} \frac{\sum_{k=1}^{N} \sum_{i=1}^{n^k} \ell_k \left(H_\theta \left(x_i^k\right), y_i^k\right)}{\sum_{k=1}^{N} n^k}. \tag{13}$$

where ℓ_k is the model utility loss of the k-th client.

D. Two-Stage Gradient Optimization

The existing federated learning work mainly uses the federated average method to aggregate the client training parameters [9], without considering the fairness problem brought by the training model of non-independent and

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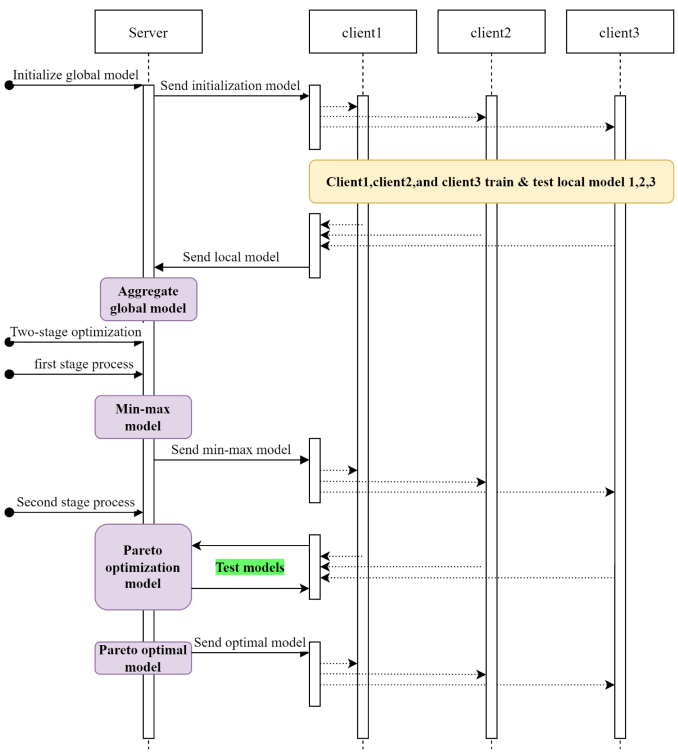

Fig. 4. Specific representation of the training and optimization model.

Algorithm 3: Two-Stage Gradient Optimization Learning.

Input: $\eta, \ell, d, g, H(\theta), W_{k,j}$
Output: $H(\theta), W_k$
Initialiation:
(a) Integrate the parameters of each SDN client $(W_{k,1}, W_{k,2}, \ldots, W_{k,j})$;
(b) Initialize the difference function g, the gradient direction d and define the utility loss ℓ of the model;
Procedure:
repeat
(a) Given a specific gradient direction d and step length η;
(b) Compute the θ^{t+1} by $\theta^{t+1} = \theta^t + \eta * d$;
(c) Find the descending direction of the k-th SDN client gradient $\bigtriangledown_\theta \ell_k$ by $\ell_k(H(\theta^{t+1})) \leq \ell_k(H(\theta^t))$;
(d) All clients gradients $G[\bigtriangledown_\theta \ell_1, \bigtriangledown_\theta \ell_2, \ldots, \bigtriangledown_\theta \ell_k]$ meet $G(d^*) \leq 0$;
(e) d^* is the direction of gradient descent and continue the iteration;
until The model H achieves Pareto optimality;
return $H(\theta), W_k$

identically distributed datasets [35], that is, the difference in algorithm decisions brought by different data sets of the client and the inconsistency of the global model learned by the client. Du et al. [36] proposed a regularization-based approach that regards the fairness of algorithms in federated learning. It is mainly to allocate a new weight value for each sample during training, which is used for the global difference between the loss target and the fair regularization processing algorithm, but it cannot explain the difference between the models of different clients. McMahan et al. [37] focused on explaining the sources of differences. Kale et al. [38] proposed a minimum maximum optimization scheme, mainly for the worst client, which may lead to the worse performance of another client. In VANETs, vehicles in SDN management subnets have high mobility, and local networks generate private network datasets differently, with differences in accuracy for different SDN clients during training, and different weights assigned to each client during aggregation by the cloud server, which reduces local model accuracy, produces inconsistent with the local training model, and creates fairness problem.

In the proposed CIDS, the different local subnets collaborate to train the model. Since the datasets within their management subnets are private datasets, which constitute differences in the data of different groups. Mathematically, we usually use population parity [39] and equal opportunity [40] to measure the difference between different groups. In this paper, we use population parity DP_k as the measure of difference. Therefore, the model difference function $g_k(H(X^k)) = DP_k$ on the client. At the same time, we use $\ell_k(H(X^k), Y^k)$ as the utility loss of the model. Under the fairness constraint, the utility of N SDN clients is optimized from the multi-objective perspective to achieve Pareto optimization.

To obtain a fair Pareto optimization, we utilize gradient based two-stage optimization method (as shown in Fig. 4) to solve the constrained multi-objective optimization problem:

1) The constrained minimum maximum optimization realizes the minimum maximum model;
2) Constrained Pareto optimization realizes Pareto optimization.

Specifically, the initial model H with given the parameters θ, we optimize H by moving to a specific gradient direction d in steps of η (as shown in Algorithm 3). If the target is lowered, $\ell_k(H(\theta^{t+1})) \leq \ell_k(H(\theta^t))$, d is the descending direction. Suppose $\bigtriangledown_\theta \ell_k$ is gradient of the k-th target,which is relative to the parameter θ. If we choose d^*, this satisfies $\bigtriangledown_\theta \ell_k(d^*) \leq 0$ for all clients. After $t+1$ iteration, d^* is a decreasing direction and $\ell_k(H(\theta^t)$ decreases. If we directly search the descending direction d to obtain $\bigtriangledown_\theta \ell_k(d^*) \leq 0$, when d is a high-dimensional vector, the calculation cost may be very large. Deside at [41] proposed to find the descending direction d in the convex shells of all target gradients, expressed as $G = [\bigtriangledown_\theta \ell_1, \bigtriangledown_\theta \ell_2, \ldots, \bigtriangledown_\theta \ell_N]$, and expressed by the following formula by searching the n-dimensional vector α^*:

$$\alpha_j \geq 0 \ (\forall j) \ and \sum_{j=0}^{N} \alpha_j = 1, \tag{14}$$

$$\bar{G} = \sum_{k=1}^{N} \alpha_k \bigtriangledown_\theta \ell_k, \tag{15}$$

$$d = \alpha^* G \in \bar{G}. \tag{16}$$

Regarding the feasibility of Pareto optimization, we define, explain and analyze the model as follows:

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1) Definition and explanation of Pareto optimization in the general case.

Pareto optimization in the general case is the problem which we find the optimal solution for our multi-objective function model.

$$V_{min} \quad f(x) = [f_1(x), f_2(x), f_3(x), \ldots, f_n(x)]^T, \tag{17}$$

$$s.t. \quad x \epsilon X, X \subseteq R^m. \tag{18}$$

where V denotes the target vector model, $s.t.$ denotes the constraint. $f(x)$ denotes n objective functions,the goal is to minimize all. x is the decision variable, $X \subseteq R^m$ is the set of constraints on its decision variables, can be explained as the range of values of the variable. Pareto optimization is the process of making a model Pareto optimal. More specifically, one model is Pareto optimal when and only when the utility of any entity can not be further optimized without degrading other entities.

2) Definition and explanation of "fairness".

Fairness refers to the difference in algorithmic decision-making between groups consisting of different data. In the proposed scheme, due to the dynamic nature in VANETs, there are large differences in the network data that result in huge differences in the algorithmic decisions.

3) Definition and explanation of Pareto optimization in programs, that is, the fairness of federated learning.

In our proposed federated learning model, after initial training, each client model needs to be optimized to find the optimal solution and reduce the variation across clients, which is our goal to learn a model H that satisfies: 1) minimum-maximum model; 2) maintains consistent performance across all clients. The multi-client model in federated learning can be defined as (1) and (2). Suppose Q is the set of all models, H is the current feasible model, $H \epsilon Q$. $\ell(H) = [\ell_1(H), \ell_2(H), \ldots, \ell_N(H)]$ denotes the utility loss vector of N learning tasks. If there is no model H' ($H' \epsilon Q$) that performs better than model H, H is a Pareto solution. Specifically, it denotes as follows:

$$\nexists\, H' \epsilon\, Q,\ s.t.\ \forall k : \ell_k(H') \leq \ell_k(H). \tag{19}$$

4) Explanation of why the method can achieve the relevant conditions in Pareto optimization.

Suppose Q is the set of all models, $\ell(H)$ is the target vector, and H denotes the current model.

1) Q^F is a set that satisfies the multi-client fairness constraint.

$$g_k(H) - \varepsilon_k \leq 0, \forall k \epsilon \{1, 2, 3, \ldots, N\}, \forall H \epsilon Q^F. \tag{20}$$

2) Q^{FM} is a set that satisfies the minimum maximum model performance, where $Q^{FM} \subset Q^F$.

$$\ell_{\max}(H') \leq \ell_{\max}(H), \forall H \epsilon Q^F, H' \epsilon Q^{FM}. \tag{21}$$

3) Q^{FP} is a Pareto optimal model set, where $Q^{FP} \subset Q^F$.

$$H' \nprec H, \forall H' \epsilon Q^F, \forall H \epsilon Q^{FP}. \tag{22}$$

4) H^* is a set that is the optimal solution model of our Pareto.

$$H^* = Q^{FP} \cap Q^{FM}. \tag{23}$$

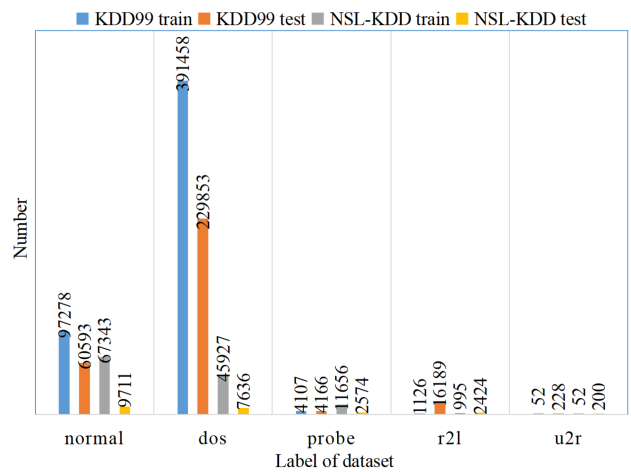

Fig. 5. KDD99 and NSL-KDD dataset label quantity description.

V. Performance Evaluation

In this section, we analyze and evaluate the performance of the proposed FFD-CIDS in terms of accuracy and performance through simulation experiments, in addition to comparing the detection model of this study with other existing schemes for comparative analysis.

A. Data Description

To verify the effectiveness of our proposal, and considering the applicability of network communication protocol in our vehicle network, we selected a popular intrusion detection dataset KDD99 [42] and its improved version NSL-KDD [43] as the experimental dataset. The dataset is extracted from network flow in a real network communication environment, involving most types of attack data in network communication, and has strong universality. There are 5 million data records in the KDD99 dataset. We take 10% of them to make the training set and test set of KDD99 experiments. NSL-KDD is an update of KDD99. It removes redundant data, has a small number, and can extract features faster through learning and training. The KDD dataset is composed of 41 features (the 1st-9th is the basic feature, the 10th-22nd is the content-based feature, the 23rd-31st is the traffic feature, and the 32nd-41st is the host-based feature) and one tag. The tag includes normal traffic and four types of attacks: Denial of Service (DoS), User to Root (U2R), Remote to Local (R2L), and Probe. Since the data set is composed of features of different data types, we also standardized and normalized the data in the dataset. The specific description is shown in Fig. 5. We mainly use different dataset distributions for experiments and divide the datasets into IID and non-IID. IID refers to the network flow data with the same data label assigned to each SDN client dataset, and non-IID refers to the network flow data with different data labels assigned to each SDN client. Suppose there are N category labels and the number of K clients.

IID: Each class of labeled data is equally distributed to each client, with each class accounting for $\frac{1}{N}$ in each client, at which point the dataset satisfies IID.

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Non-IID: The data of N category labels are *randomly* assigned to K clients, at which point the dataset satisfies Non-IID.

B. Experimental Setup

In this study, we use Python and PyTorch1.6 to implement our proposed federated collaborative intrusion detection system on a computer-based CPU. Scikit-learn is used for the traditional model and PyTorch is used for the deep learning model. The computer uses Intel (R) core (TM) i5-10500 CPU and 8 GB RAM. In our experiment, we simulated one cloud server and several SDN clients in the vehicle networking communication model. The SDN client uses its private data set to train and deploy the model, and the cloud server aggregation optimization model.

C. Performance Metrics

In our experiment, we considered several basic indicators, including True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). TP refers to the number of attack traffic correctly predicted, TN refers to the number of normal traffic correctly predicted, FP refers to the number of attack traffic incorrectly predicted, and FN refers to the number of normal traffic incorrectly predicted. In a recent study, Mehedi et al. [44] proposed intrusion detection research based on deep learning method, provides a comprehensive performance analysis method, and emphasizes model performance analysis, so as to better compare and analyze model performance. Convergence time can measure model time complexity, Accuracy can measure classification performance, Precision can measure positive samples predicted correctly, recall can measure how much can be predicted, F1-score can measure the balance point between Precision and recall, and AUC can measure classifier performance. Between 0 and 1, the closer to 1, the stronger the classifier. According to the above suggestions, we selected *Accuracy*, *Precision*, *Recall*, *F1-score*, and *AUC* as the evaluation criteria.

Accuracy refers to the ratio of correct prediction to all predictions, expressed as:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}. \tag{24}$$

Precison refers to the ratio of the number of positive cases correctly predicted to the total number of positive cases predicted, expressed as:

$$Precision = \frac{TP}{TP + FP}. \tag{25}$$

Recall refers to the ratio of the number of correctly predicted positive cases to the total number of actual positive cases, expressed as:

$$Recall = \frac{TP}{TP + FN}. \tag{26}$$

F1-score refers to the sum of precision and recall is closer to the smaller of the two numbers, so the value of F1-score is greatest when the precision and recall are close, expressed as:

$$\frac{2}{F1 - score} = \frac{1}{Precision} + \frac{1}{Recall}. \tag{27}$$

AUC represents the area under the ROC curve, and the relationship between the recall rate and the false positive rate is plotted according to different classification thresholds.

$$FPR = \frac{FP}{FP + TN}. \tag{28}$$

D. The Performance Analysis of Proposed Collaborative Intrusion Detection Model

To evaluate the performance of the proposed model, we conducted experiments on KDD99 and NSL-KDD datasets. According to the proposal of [45], AI methods are trained with 80% of the dataset as the training set and 20% as the hyperparameter optimization and validation training set. Meanwhile, to avoid overfitting problems, this allocation ratio is referred to as the optimal ratio [46]. Specifically, we simulate the Cloud Server and three SDN clients in VANETs. Each SDN controller uses a specific initial non-IID dataset to train the intrusion detection model. In our experiments, both the KDD99 dataset and the NSL-KDD dataset have five major categories of labels and three numbers of clients. To meet the characteristics of the current complex and variable network communication data in VANET as much as possible, we first divided these five categories of data in different proportions, the first category: 0.3, 0.3, 0.4, the second category: 0.2, 0.2, 0.6, the third category: 0.1, 0.8, 0.1, the fourth category: 0.4, 0.4, 0.2, and the fifth category: 0.5, 0.4, 0.1. And then the three clients follow the divided data as their training dataset. The number of KDD99 training datasets is 9675, 39353, and 444993 respectively, and the number of NSL-KDD training datasets is 95384, 27964, and 2625 respectively. At the same time, each training set is data with different numbers and labels. The specific Non-IID partitioning details and results about the dataset are shown in Fig. 6. As proposed experimentally by Gong et al. in [47] the important impact of Non-IID datasets on federated learning performance models, there is a great correlation between the distribution of datasets and model performance, as well as better validation of a model's performance. In Fig. 6, we have a large variability in the distribution of the datasets, not only in the number of differences but also in the type of data labels. The evaluation results of the proposed model are shown in Table II. The model proposed by us performs well on different SDN clients, and there is no problem that the training models of local SDN clients differ greatly due to the difference in private data. Due to the gradient-based optimization algorithm in the cloud server. The detection accuracy of the KDD99 dataset is higher, which is that because there are more data in the KDD99 dataset, which improves the training accuracy. In addition, the AUC of the proposed model is close to 1, which indicates that our model has a higher classification performance. Wang et al. [48] constructed an anomaly detection model

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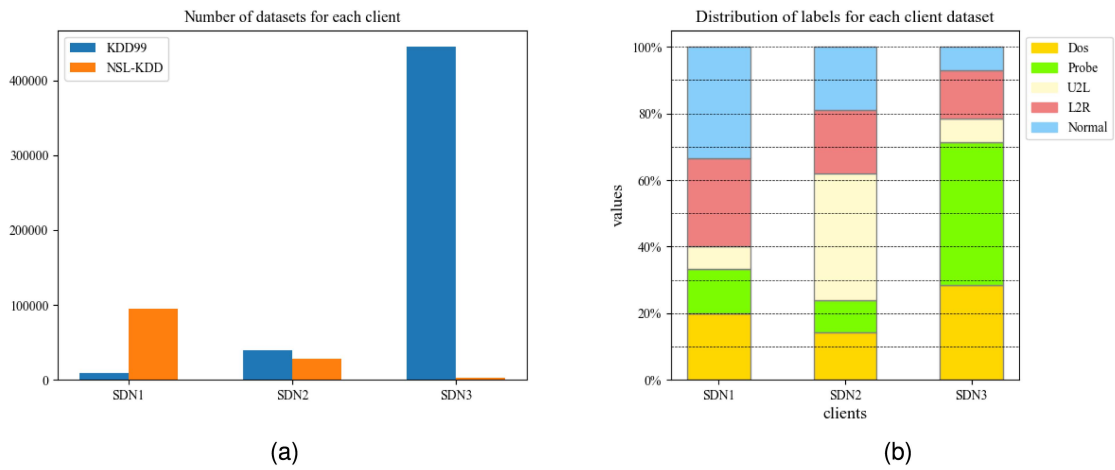

Fig. 6. In the figure, the proportion of datasets randomly divided into Non-IID and the number of datasets in each client after the division, show the specific details of the experimental datasets, which are helpful to better detect the model performance. (a) Graph of the number of client datasets. (b) Non-IID dataset partitioning scale graph.

TABLE II
KDD99 AND NSL-KDD DATASET RESULT IN EFFECTIVENESS

		KDD99			NSL-KDD		
		Accuracy	Auc	F1-score	Accuracy	Auc	F1-score
FFD-CIDS	SDN1	0.9902	0.9948	0.9930	0.9587	0.9847	0.9890
	SDN2	0.9898	0.9946	0.9913	0.9604	0.9867	0.9905
	SDN3	0.9912	0.9947	0.9920	0.9611	0.9865	0.9882

based on a hierarchical federated deep reinforcement learning algorithm to enhance the privacy and security of the device anomaly detection model, and experimentally demonstrated the reward mechanism of deep reinforcement learning, which is beneficial to relieve local model discrepancies and improve security. Our FFD-CIDS approach ensures the effectiveness of individual client training by deep learning convolutional neural network algorithms for local model training. In addition, to relieve the differences in the intrusion detection models of individual SDN servers the method passes a gradient-based two-stage optimization learning algorithm to ensure the local performance while gradually reducing the model differences between individual clients. In KDD99 dataset, the training process of our algorithm and the variation of its important performance metrics are shown in Fig. 7. Two recent studies have proposed new design solutions to improve intrusion detection performance and accelerate model training efficiency. The former constructs an interpretable algorithm to improve algorithm stability [14], and the latter improves the efficiency problem of machine learning algorithms through hyperparametric optimization algorithms [29]. For the artificial intelligence method, the efficiency of the algorithm model is reflected by the speed of change of the loss value. When Epoch = 80, the Loss value almost stops changing and stabilizes, which shows that this model can converge quickly. When Loss remains stable at lower values, Accuracy also maintains high accuracy, which indicates the stability of our proposed intrusion detection model. In addition, the F1-score is close to 1, which shows the higher quality of the model.

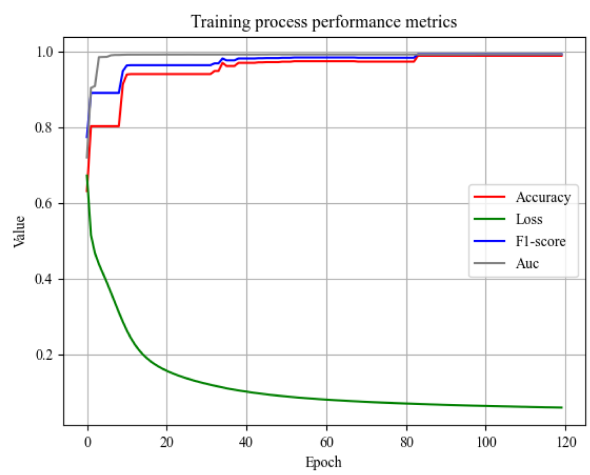

Fig. 7. Performance metrics for the FFD-CIDS training process.

E. Comparative Experiments of the Existing VANET-Based IDS Methods

To prove the performance difference between the proposed model and the latest methods, we use the KDD99 dataset to compare its performance with some of the most advanced schemes and traditional schemes. The experimental results of

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TABLE III
RESULTS OF DIFFERENT METHODS IN EFFECTIVENESS

Reference	FL	ML	DL	Collaborative	Accuracy
CIDS-SDN[24]			√	√	0.9830
FL-Block[26]	√			√	0.94
SP-CIDS[30]			√	√	0.9694
MLP[49]		√			0.91
SVM		√			0.94
RF		√			0.9786
FFD-CIDS	√		√	√	0.9904

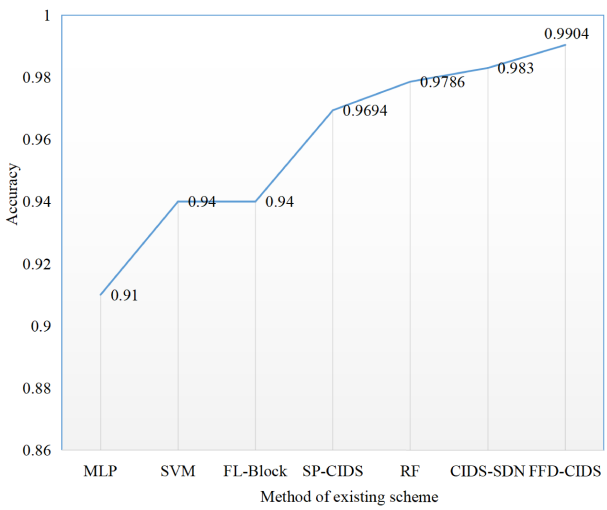

Fig. 8. Comparison of detection accuracy of existing schemes.

these schemes are shown in Table III and Fig. 8, including the deep learning generation countermeasure network model [24], the federated learning and blockchain integration model [26], the security and privacy distributed model [30], the multilayer perceptron model [49], the support vector machine algorithm model and the random forest algorithm model. We choose these works because they are the intrusion detection models that are compared in the field of the VANETs, and involve deep learning, machine learning, and federated learning. It can be seen from Table III that the results of the method based on the deep learning model are better than the traditional machine learning method because more features can be processed in the training process of the deep learning model. In addition, traditional machine learning algorithms have high time costs, low efficiency, and poor performance. The deep learning model can improve detection performance and improve detection accuracy. The federated deep learning model algorithm proposed in this paper has the best accuracy performance, and the application of federated learning reduces the time cost of algorithm training, improves detection efficiency, and ensures data privacy to a certain extent. At the same time, it can be seen from Table III that the cooperative intrusion detection system model (including our model, the deep learning generation countermeasure network model [24] and the security and privacy distributed model [30]) has higher detection accuracy.

TABLE IV
RESULTS OF SINGLE SDN CLIENT IN EFFECTIVENESS

		Accuracy	Auc
Centralized	SDN1	0.7753	0.8758
	SDN2	0.7690	0.8636
	SDN3	0.7691	0.8977
FFD-CIDS	SDN1	0.9902	0.9948
	SDN2	0.9898	0.9946
	SDN3	0.9912	0.9947

F. Single SDN Client Centralized Learning Model

We utilize multiple SDN controllers as the client cooperation to train the federated learning model in SDVN. In order to prove the effectiveness of the cooperation between multiple clients, we made three experiments using KDD99 dataset, and let each SDN client independently train the intrusion detection model based on local dataset. The experimental results are shown in Table IV. We can see from the table that the detection accuracy of a single SDN client training model is only 76%, which is far lower than the collaborative intrusion detection model proposed by us. A single centralized intrusion detector has poor expansibility, takes a long time to train, and requires a large amount of data resources. Therefore, the intrusion detection system model proposed in this paper can improve the detection performance by cooperating among multiple SDN clients, and has significant utility and occupies less resources. The distributed collaborative intrusion detection model has better application prospects than the centralized intrusion detection model.

G. Federated Average Aggregation Algorithm

Federated learning algorithms usually use the federated average (FedAvg) to aggregate local models. FedAvg aggregation means that during the training process, the server will average the weights and gradient parameters received from all clients to speed up the model training and to obtain the global model effect. To verify the effectiveness of the gradient-based optimization algorithm used in the cloud server in this paper, we made an experimental comparison with the federal average algorithm on the NSL-KDD dataset. The experimental results are shown in Table V. It can be seen from the table that the accuracy of the federated average aggregation algorithm can reach 91%,

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TABLE V
RESULTS OF FEDAVG AGGREGATION MODEL IN EFFECTIVENESS

		Accuracy	Auc
FedAvg	SDN1	0.9156	0.9839
	SDN2	0.9219	0.9824
	SDN3	0.9206	0.9825
FFD-CIDS	SDN1	0.9681	0.9937
	SDN2	0.9675	0.9934
	SDN3	0.9672	0.9933

TABLE VI
RESULTS OF SDN CLIENT NUMBER AND DATASET DISTRIBUTION IN EFFECTIVENESS

		IID	non-IID
K=3	**Accuracy**	0.9680	0.9676
	Auc	0.9947	0.9950
K=5	**Accuracy**	0.9681	0.9681
	Auc	0.9956	0.9968
K=10	**Accuracy**	0.9620	0.9614
	Auc	0.9982	0.9975

while the federated deep learning model aggregated by the optimization algorithm can have 98% accuracy. Therefore, the gradient optimization algorithm we use has better performance than the federated average. Note that it can be seen from Tables IV and V that using the federated learning framework algorithm can effectively improve the performance of IDS.

Finally, in real-world scenarios of VANETs, where vehicles are highly mobile and widespread, the scalability of CIDS is an important issue as the scope of VANETs increases. We consider the impact that both the number of clients and different data distributions have on the CIDS model and conduct experimental validation by utilizing the NSL-KDD dataset. Specifically, we respectively set the number of SDN clients to K = 3, 5, and 10, each of which conducts non-IID and IID private dataset experiments, and record the corresponding experimental results. The experimental results are shown in Table VI, and the visualization accuracy is shown in Fig. 9. We observe that as the number of SDN clients increases, the detection accuracy of our proposed CIDS model remains almost constant and keeps high stability. It can be seen that our proposed gradient optimization-based federated deep learning intrusion detection model has good scalability and robustness and will not lead to degradation in the performance of the CIDS model due to increasing the number of SDN controllers or increasing the scale and scope of the VANET. Meanwhile, the experimental results of our CIDS model on IID and non-IID data distributions are almost the same. This is because our two-stage gradient optimization algorithm plays an important role in cloud server aggregation, especially for data resources with non-IID data distribution. In addition, as the scale of VANET increases, the number of SDN controllers also increases, in which case the FFD-CIDS training model overhead increases appropriately, and this overhead is unavoidable. However, this overhead is controllable because the

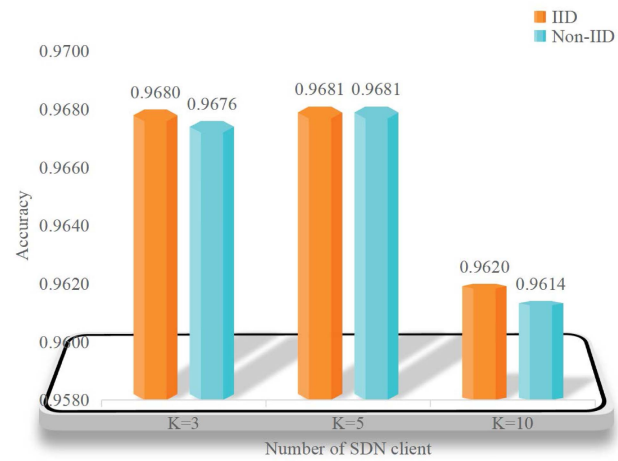

Fig. 9. Experimental results of client number and data distribution.

increase in SDN controllers means that the overhead of local SDN controller training is reduced, which is more beneficial for local SDNs to manage the vehicles within their range. The increased overhead of the cloud server aggregation optimization model is well mitigated by the current high-performance servers with superior computing power. Therefore, this overhead is reasonable.

VI. CONCLUSION

To quickly and efficiently detect network intrusion and attacks in VANETs, this paper proposes a new collaborative intrusion detection model for the SDVN based on the federated deep learning framework. The use of FL ensures the training efficiency of the model and the privacy of the local VANET communication data. To address the inconsistency problem caused by the difference in the distribution of private data resources of local clients, this study explores a two-stage gradient optimization algorithm from the perspective of multi-objective optimization so that the client training model can obtain the Pareto optimal solution and can avoid the deterioration in detection performance owing to the difference in local vehicle network traffic. The collaborative intrusion detection model proposed herein is based on the entire vehicle network and not on the local vehicle network; hence, it can solve the problem of bias current. Rigorous experimental simulation results verify the validity of detection in IID and non-IID datasets. This study also compares the proposed method with the latest and traditional methods, and the proposed method is found to present the best performance. Considering the current lack of data resources in VANETs, future research will need to introduce semi-supervised learning into VANETs based on this new CIDS model to address the problems of data labeling and datasets.

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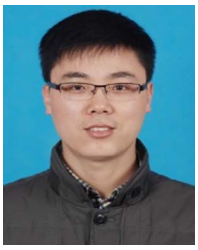

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